Total No. of Pages 2 Roll No.....

VIth SEMESTER B.Tech.(Computer)

End Semester Examination(supplementary)August 2018

co304 Artificial Intelligence

Paper Code Title of the subject

Time:3 Hours Max. Marks: 50

Note:	Answer any 5 questions. Assume suitable missing data, if any.

Q1. (a) What are the advantages and disadvantages of intelligent search over non intelligent search? Give an example of intelligent and non intelligent program for Tic Tac Toe problem.

(b) Define state space and give the state space for water jug problem. Define the production rules and work out the production rule for this problem.

(5,5)

Q2. (a) Explain with the help of suitable search tree the search procedure best first search and A* search.

(b) Write the algorithm for AO* Search and differentiate it from A* Search

(5,5)

Q3. (a) With suitable example illustrate how you solve crypt arithmetic problem.

(b) Explain Minimax Procedure and show its step for Tic Tac Toe problem

(5,5)

Q4. (a) Explain the different steps of converting wff in to clause form with the help of suitable examples.

(b) Explain the steps of resolution in the form of flow graph. Work out these steps to infer that Jack likes mangoes from following given facts;

i. John Likes all kind of food
ii. Apples are food
iii. Any thing one eats and is not killed by is a food
iv. Bill eats mangoes and is still alive

(5,5)

Q5. (a) What are the main constituents of knowledge representation techniques: Semantic Nets.? How are they represented ?, Give the required inference procedures to infer new facts.

(b) What are the limitations in semantic nets. How are they overcome in frames. Give a general structure of frame and its declaration in a programming language. How inference is done in frames.

(5,5)

Q6. (a) Explain the block diagram to design Expert System.

(b) Explain the different techniques to handle non monotonic reasoning

(5,5)

Q7. Write short notes on **any two** from followings:

(a) Probabilistic Reasoning
(b) Prolog and knowledge Representation.
(c) Syntactic Analysis and Semantic Analysis for NLP
(d) Learning using neural Nets
(e) Fuzzy Logic and Uncertainty

(5,5)